Energy Procurement **Optimization**

Multi-Objective Linear Programming (MOLP) Approach

Decision Tree

Linear Programming

Green Energy

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Executive Summary

Objective

A manufacturing plant must procure electricity every hour to meet production demand.

Determine the **optimal hourly portfolio mix** – how much electricity to procure from each source – to balance simultaneously:

- **(1) minimize total cost;**
- **(2) minimize supply risk;** and
- **(3) Minimizing CO_2 intensity.**

Context & Challenges

No single source satisfies all three sub-objectives. The optimal mix depends on prices, demand levels, and how much the company values sustainability.

Three available sources:

	Cost	Supply Reliability	CO_2 intensity
(1) Standard Contract	Medium	Generator failure possible	Upon hourly grid mix
(2) RE Certified Contract	High	Generator failure possible	0
(3) Spot Market	Low (avg)	Higher if reported timely to operator	Upon hourly grid mix

Proposed Model

Scenario-based MOLP with:

(1) Decision tree structures demand & price scenarios; and

(2) Multi-objective Linear Programming solves optimal portfolio allocation in expectation across all scenarios.

Pipeline

1 – Decision Framework
Define scenarios & MOLP problem structure

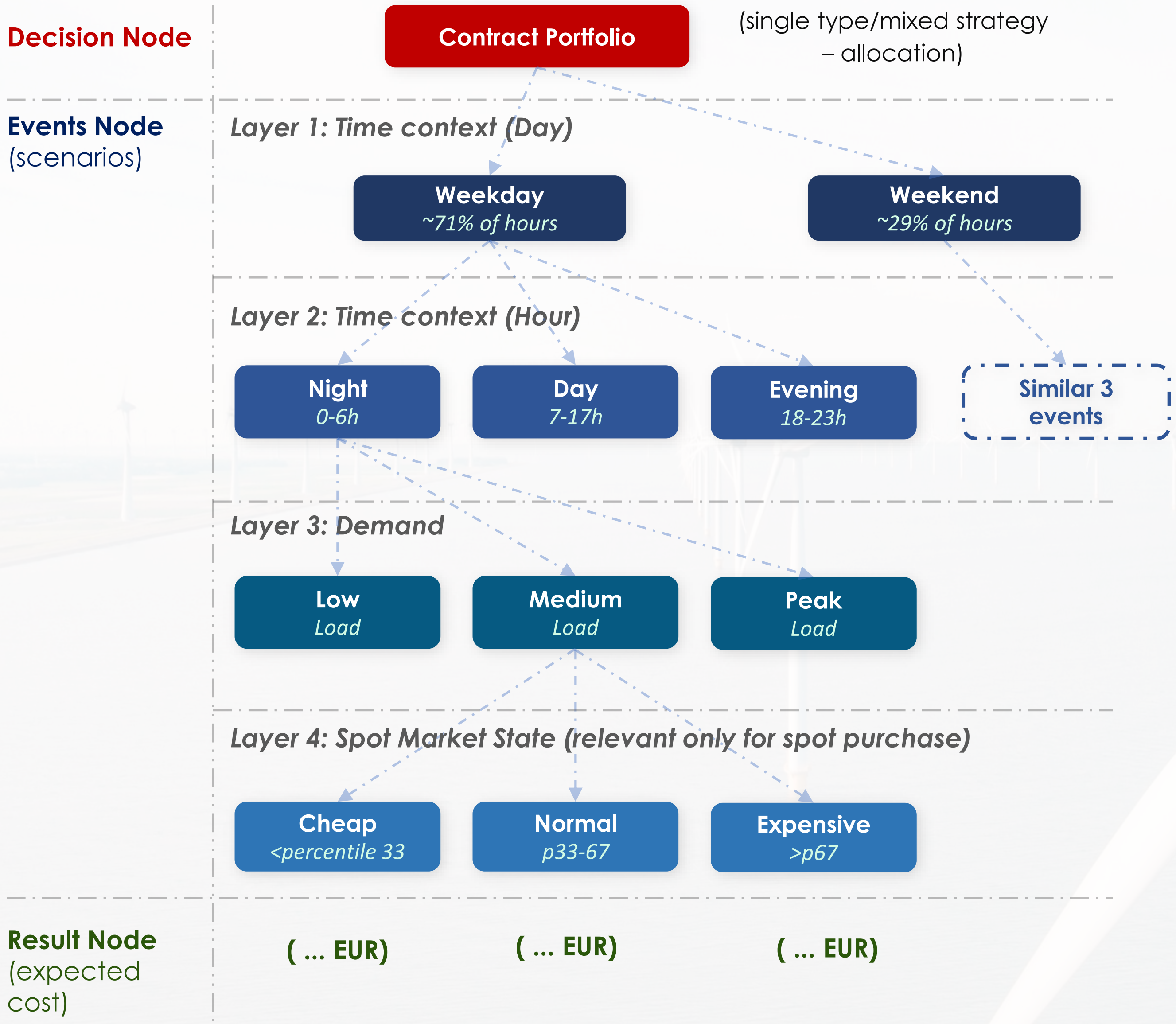
2 – Data Collection
Obtain firm and electricity market data.

3 – Scenario Building
Estimate probability for decision trees from data.

4 – Solve & Optimize
LP solver per scenario to obtain optimal portfolio.

5 – Sensitivity Analysis
Across different weights/contract premium.

Decision Tree



Scenario Structuring

- **Probabilities of each scenario are estimated from data:**
  - Demand tiers → Load_Type frequency in DAEWOO dataset
  - Price tiers → Percentile splits (P33 / P67) from ENTSO-E Finland spot prices
  - Time context → Weekday/weekend & hour distributions from data
- Each leaf node probability = product of conditional probabilities along its path.
- **Total scenarios:**
  - If spot prices involved: up to 54 leaf nodes (2 × 3 × 3 × 3)
  - Fixed contracts only: 1 leaf node

Optimization Model – Objectives

Aggregate Objective Function

$$\min_{x_1(t),\, x_2(t),\, x_3(t)} \sum_{s} p(s).\,[w_1.\,\mathrm{Cost}(s) + w_2.\,\mathrm{Supply\ Risk}(s) + w_3.\,CO_2\ \mathrm{Intensity}(s)]$$

$x_1(t)$ $x_2(t)$ $x_3(t)$ **Decision Variables (solved by LP)** – MWh bought from Standard Contract/ RE-Certified Contract/ Spot Market

w_1 w_2 w_3 **Priority Weights (set by the company)** – How much company values cost savings/ supply reliability/ green score.

s p(s) **scenario s** (combination of time context, demand level, spot price state) & **probability p(s)** of each scenario

① Cost $= \sum_t \sum_{i=1}^{3} p_i(t).\,x_i(t)$

where:

- p_1 standard contract price = E[spot] × 1.10
- p_2 RE contract price = E[spot] × 1.15
- $p_3(t)$ spot price at hour t

Multiply price [€/MWh] × quantity [MWh] → cost [€]

Note: t = each hour in planning horizon (8,760 hrs/year)

② Supply Risk $= \sum_t \mathrm{short\_fall}(t).\,\mathrm{p\_bal}$

where:

short_fall (t) = max(0, ϵ.($x_1 + x_2$)): possible amount that contract generator fails to deliver

ϵ: fraction of committed amount contract generator fails to deliver

p_bal : price to buy from balancing market

Captures production loss from unmet demand

Assume baseline **ϵ(t) = 0.02,**

③ CO_2 Ints. $= \sum_t (x_1(t) + x_2(t)).\,ci(t)$

where:

ci(t): grid carbon intensity, varies by hour

Unit: gCO_2

Optimization Model – Programming

The **LP solves for $x_1(t)$, $x_2(t)$, $x_3(t)$** that **minimize the probability-weighted sum of all three objectives** across all defined scenarios.

Constraints:

	Inequality/Equation	Explanation
(1) Demand Coverage	$\mathbf{x_1(t) + x_2(t) + x_3(t) = D(t)}$	Must cover demand at each hour
(2) Standard Contract Cap	$\mathbf{x_1(t) \leq \overline{C_1}}$	Cannot exceed agreed contract capacity
(3) RE Contract Cap	$\mathbf{x_2(t) \leq \overline{C_2}}$	Cannot exceed agreed contract capacity
(4) Non-negativity	$\mathbf{x_1(t),\ x_2(t),\ x_3(t)\ \geq\ 0}$	Cannot procure negative quantities

* *Note: Assume* $\overline{C_1}, \overline{C_2} = \mathbf{50\%, 20\%}$ of average $\mathbf{D(t)}$

Expected Outputs

01
Scenario Probability Table
Cross-tabulated probabilities for all leaf nodes, derived from data.

02
Optimal Portfolio Mix
Hourly x_1 / x_2 / x_3 allocation per scenario.

03
Cost vs Baseline
Expected savings vs naive always-spot strategy, quantified in €/year.

04
Trade-off Frontier
How optimal cost and green score shift as w_1, w_2, w_3 weights change.

05
Contract Premium Sensitivity
Optimal mix at 5% / 10% / 15% premium – value of contract stability.

06
Supply Reliability Analysis
Impact of contract generators' failure to deliver on cost/shortfall